

PRACTICAL – 9

CODE:-

```
#include<stdio.h>

int main()
{
    int ref[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3] = {-1,-1,-1};
    int n = 12;
    int i,j,k,pos;
    int hit = 0, fault = 0;
    int found;
    printf("\n===== OPTIMAL PAGE REPLACEMENT =====\n");
    printf("\nReference String : ");
    for(i=0;i<n;i++)
        printf("%d ", ref[i]);
    printf("\n\nFrame Status:\n");
    for(i=0;i<n;i++)
    {
        found = 0;
        // check HIT
        for(j=0;j<3;j++)
        {
            if(frames[j] == ref[i])
            {
                hit++;
                found = 1;
                printf("\nPage %d -> HIT | ", ref[i]);
            }
        }
    }
}
```

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        break;
    }
}

// PAGE FAULT
if(found == 0)
{
    int farthest = -1;
    pos = -1;
    for(j=0;j<3;j++)
    {
        int next = -1;
        for(k=i+1;k<n;k++)
        {
            if(frames[j] == ref[k])
            {
                next = k;
                break;
            }
        }
        if(next == -1)
        {
            pos = j;
            break;
        }
        if(next > farthest)
        {
            farthest = next;
            pos = j;
        }
    }
}

```

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    }

    frames[pos] = ref[i];

    fault++;

    printf("\nPage %d -> FAULT | ", ref[i]);

}

for(j=0;j<3;j++)

    printf("%d ", frames[j]);

}

printf("\n\nTotal References = %d", n);
printf("\nTotal Page Hits = %d", hit);
printf("\nTotal Page Faults = %d", fault);


float hitp = (float)hit/n * 100;
float faultp = (float)fault/n * 100;


printf("\nHit Percentage = %.2f%%", hitp);
printf("\nFault Percentage = %.2f%%", faultp);


return 0;

}

```

OUTPUT -

Output

===== OPTIMAL PAGE REPLACEMENT =====

Reference String : 1 2 3 4 1 2 5 1 2 3 4 5

Frame Status:

Page 1 -> FAULT | 1 -1 -1

Page 2 -> FAULT | 1 2 -1

Page 3 -> FAULT | 1 2 3

Page 4 -> FAULT | 1 2 4

Page 1 -> HIT | 1 2 4

Page 2 -> HIT | 1 2 4

Page 5 -> FAULT | 1 2 5

Page 1 -> HIT | 1 2 5

Page 2 -> HIT | 1 2 5

Page 3 -> FAULT | 3 2 5

Page 4 -> FAULT | 4 2 5

Page 5 -> HIT | 4 2 5

Total References = 12

Total Page Hits = 5

Total Page Faults = 7

Hit Percentage = 41.67%

Fault Percentage = 58.33%

=== Code Execution Successful ===